

VOCABULARY BUILDER

Radical Alternatives to the Economy – Key Words and Phrases • LEVEL C

The words and phrases below appear in the article *"The Economy Is an Emergent Phenomenon, in Which Humans Are the Building Blocks"*, a Features interview with complexity theorist J. Doyne Farmer conducted by Thomas Lewton, published in *New Scientist* (20 April 2024). The vocabulary spans three registers: economic theory, scientific methodology, and business and policy analysis. Study the table carefully before completing the gap-fill exercise in Part 2.

PART 1 – VOCABULARY TABLE

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
emergent phenomenon	a property or behaviour that arises from a complex system but cannot be predicted from its individual parts alone	<i>The economy is an emergent phenomenon — its booms and crashes cannot be explained by looking at individual buyers and sellers in isolation.</i>
complexity economics	an approach that treats the economy as a complex adaptive system, modelling how many agents interact rather than assuming rational equilibrium	<i>Complexity economics uses tools from biology and physics to model how real economies evolve and behave.</i>
mainstream economics	the dominant set of economic theories and methods taught in most universities and used by most governments and institutions	<i>Farmer argues that mainstream economics failed to predict the 2008 financial crisis because its models were too simplistic.</i>
pseudoscience	a set of beliefs or methods that claims to be scientific but lacks the rigour, testability, or predictive power of genuine science	<i>Critics have long accused economics of being a pseudoscience because its mathematical models so often fail in practice.</i>
utility function	in economics, a mathematical representation of a person's preferences — what they like and dislike — used to model decision-making	<i>Standard economic theory assumes that people make decisions to maximise their utility function, but this ignores how humans actually behave.</i>
Homo economicus	the theoretical model of a perfectly rational human being who always makes decisions to maximise their own self-interest	<i>The concept of Homo economicus dominated economics from the 1970s until researchers began showing that real people rarely behave this way.</i>
equilibrium	a stable state in a system in which opposing forces are balanced and there is no tendency for change	<i>Standard economics assumes markets will reach equilibrium, but complex systems often fluctuate without ever settling.</i>
austerity	a policy of reducing government spending, typically in response to debt, often with negative social consequences	<i>Farmer cites the Greek austerity crisis as an example of economists disagreeing violently on the right course of action.</i>

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
agent-based model	a computer simulation in which many individual 'agents' follow simple rules and interact, producing complex system-wide behaviour	<i>Using an agent-based model, Farmer's team predicted a 21.5% GDP drop for the UK during COVID-19 before the outcome was known.</i>
verisimilitude	the appearance of being true or real; the quality that makes a model believable and realistic	<i>Agent-based models aim for verisimilitude by modelling how people actually make decisions, not how theory says they should.</i>
emergent behaviour	actions or patterns that arise from the interaction of many simple components rather than being built into any one of them	<i>Ant colonies exhibit emergent behaviour — mushroom-farming and aphid-ranching arise from millions of individual insects following simple rules.</i>
equilibrium economics	the branch of mainstream economics that assumes markets naturally tend towards a stable, balanced state	<i>Equilibrium economics struggles to explain financial bubbles and crashes, which require non-equilibrium thinking.</i>
intractable	too difficult or complex to solve or deal with using conventional methods	<i>Climate change and financial instability are intractable problems for standard economic models, which is where complexity economics offers an alternative.</i>
no-growth economics	a school of thought arguing that economies should stop pursuing GDP growth in order to protect the environment and human wellbeing	<i>Farmer is not a fan of no-growth economics, arguing that growth is needed to fund the transition to sustainable energy.</i>
lambasted	strongly and publicly criticised or attacked	<i>Economics is often lambasted for being a pseudoscience, given its poor record of predicting major events.</i>
belie	to give a false impression of something; to contradict or misrepresent	<i>The dense mathematical formulae in economic models belie their subjectivity — they look rigorous but rest on questionable assumptions.</i>
quantitative predictions	specific, measurable forecasts that can be tested against real-world outcomes	<i>Complexity economics can now make quantitative predictions — such as next quarter's GDP — rather than just qualitative descriptions.</i>
proof of principle	a demonstration that a method or idea is feasible and capable of producing the expected results	<i>Accurately predicting COVID-19's economic impact before the data was available provided a powerful proof of principle for agent-based modelling.</i>
network effects	the indirect consequences that arise when many interconnected agents or systems influence each other	<i>When BP divests from oil, the effects ripple through labour markets, suppliers, and energy prices via complex network effects.</i>
divest	to sell off assets or investments, often for ethical or strategic reasons	<i>Companies that divest from fossil fuels face short-term losses but may gain long-term reputational and regulatory advantages.</i>

WORD / PHRASE	DEFINITION	EXAMPLE SENTENCE
policy-makers	people in government or institutions who decide on the rules and strategies that shape public life	<i>Farmer argues it is time for policy-makers to take complexity economics seriously as a tool for navigating major transitions.</i>
profit motive	the incentive to act in order to earn financial gain, which drives much business and economic behaviour	<i>Farmer argues that harnessing the profit motive — rather than fighting it — is the most realistic way to accelerate the green transition.</i>
harness	to control and make use of a natural force or human tendency for a productive purpose	<i>If we can harness the profit motive and constrain it with good regulation, we can make the transition to clean energy far faster.</i>
transition	the process of changing from one state or system to another, used here for the shift to sustainable energy	<i>Complexity economics can model how workers will transition to different occupations as the energy system changes.</i>

PART 2 – GAP-FILL EXERCISE

Complete the sentences using the correct word or phrase from the word bank below. You may need to change the form of the word. Most sentences require more than one answer. Several sentences test your understanding of the article's central argument — students who have read carefully will have a clear advantage.

WORD BANK

emergent phenomenon	complexity economics	mainstream economics	pseudoscience
utility function	Homo economicus	equilibrium	austerity
agent-based model	verisimilitude	emergent behaviour	equilibrium economics
intractable	no-growth economics	lambasted	belie
quantitative predictions	proof of principle	network effects	divest
policy-makers	profit motive	harness	transition

1	Economics is often _____ for being a _____ — its dense mathematical formulae _____ its subjectivity, and it has a poor track record of making accurate predictions.
2	The core problem with _____ is its reliance on _____ — the idea that people are perfectly rational agents who always act to maximise their own _____.
3	Standard _____ assumes that markets tend towards _____, but this is only a decent approximation in very simple cases and breaks down entirely in complex, fast-moving economies.
4	Complexity economics treats the economy as an _____ — a system that exhibits _____ which cannot be found in its individual parts, just as ant colonies exhibit behaviours no single ant possesses.

5	The models used in _____ are _____, meaning that individual 'agents' follow simple rules and interact, producing realistic system-wide behaviour that emerges from the bottom up.
6	A key requirement for good economic models is _____ — they must follow the same basic principles that the real world does, not just produce mathematically elegant but unrealistic results.
7	Farmer's team made accurate _____ about the UK economy during COVID-19 — including GDP, unemployment, and industry-level impacts — providing a compelling _____ for their approach.
8	When Farmer asks what happens if BP decides to _____ from oil and invest in solar farms, the answers are not obvious because of all the indirect _____ running through the economy.
9	The _____ crisis after 2008 is a perfect example of economists giving completely contradictory advice — some demanded _____, while Nobel Prize-winners argued for the opposite.
10	Farmer is sceptical of _____ economics, arguing that we need growth to fund the _____ quickly — and that pushing a no-growth agenda risks alienating people from climate action entirely.
11	The most productive narrative, Farmer argues, is that climate change is a business opportunity rather than a burden — if we can _____ the _____ and constrain it with good policy, we can build a fairer economy.
12	For _____ to take complexity economics seriously, it must demonstrate that it can make reliable _____ — not just describe the past, but genuinely forecast what comes next.
13	Standard economics treats people as _____ — rational maximisers of their _____ — but this was shown to be a poor model of actual human behaviour by the early 2000s.
14	The economy is, in Farmer's words, an _____ in which humans are the building blocks — and understanding it requires treating it as a complex system, not a set of mathematical equations.
15	Complexity economics borrows tools from fields like biology and physics, applying them to understand how billions of specialised human _____ interact and what _____ from all of those interactions.

Write your answers here:

1. _____	2. _____	3. _____
4. _____	5. _____	6. _____
7. _____	8. _____	9. _____
10. _____	11. _____	12. _____
13. _____	14. _____	15. _____